

Based on the absolute latest 2024 and 2025 IEEE research papers, our original architecture was highly accurate, but we can significantly upgrade it. Here is the final, institutional-grade model architecture I am proposing for our Multi-Branch Deep Learning Trading System:

1. Data Engineering & Preprocessing Layer

Before data reaches our neural networks, we must optimize the pipeline for speed and visual feature extraction.

- **Data Format:** We are transitioning all historical data into TFRecords to maximize GPU utilization and prevent CPU bottlenecks during training.
- **C++ Optimization:** When parsing these TFRecords, we must explicitly avoid wrapping `tf.io.parse_single_example` with `tf.py_function`. Doing so forces TensorFlow out of its highly optimized C++ environment into Python, which will severely degrade our pipeline's speed.
- **Image Encoding (GAF):** For our CNN, we will encode our 1-minute and 5-minute time-series data into 2D images using the Gramian Angular Field (GAF). This technique preserves absolute temporal relations by representing the data in a polar coordinate system.
- **Feature Selection (CULR):** Instead of standard OHLC (Open, High, Low, Close) prices, we will use CULR (Close, Upper shadow, Lower shadow, Real-body). Research proves that OHLC creates images that are too visually similar, whereas CULR generates distinct visual contrast that pushes our CNN pattern recognition accuracy above 92%.

2. Branch A: The Tactical Branch (Short-Term Execution)

- **The Final Model:** HCLA (Hybrid CNN-LSTM with Attention).
- **How it works:** This 2025 architecture officially replaces our standalone CNN + GRU approach. It utilizes a CNN to extract spatial features, an LSTM for temporal learning, and an Attention Mechanism.
- **Attention-weighted Hybrid Feature Extraction (AHFE):** Instead of treating all data points equally, the attention layer dynamically focuses only on the most significant, volatile time steps in our order book data.
- **Dynamic Error Feedback Adjustment (DEFA):** We are introducing an adaptive correction mechanism that actively refines the model's live predictions based on the recent prediction errors it just made in the live market.

3. Branch B: The Strategic Branch (Context & Forecasting)

- **The Final Model:** TFT (Temporal Fusion Transformer) with Quantile Loss.
- **How it works:** The TFT remains our heavy hitter for handling static metadata (like sector market cap, or days to expiry) alongside past stock prices.
- **Key Upgrade:** We must implement a Quantile Loss Function. Instead of forcing the TFT to predict a single absolute price, this function allows the model to produce Probabilistic Forecasts (e.g., 10%, 50%, and 90% confidence intervals). This explicitly quantifies market uncertainty, giving our Fuzzy Logic Risk Manager the exact mathematical boundaries needed to size our positions.

4. Branch C: The Memory Branch (Long-Term Trend)

- **The Final Model:** TLSTM (Transductive LSTM) or a Hybrid LSTM-Transformer.

- **How it works:** Standard LSTMs apply a rigid global rule based on old training data, which we need to move past.
- **TLSTM Implementation:** By using a TLSTM, our model will dynamically modulate its parameters based on how similar the current live tick is to recent data, giving higher priority to immediate temporal patterns.
- **Hybrid Alternative:** Alternatively, a Hybrid LSTM-Transformer integrates the localized sequential learning of the LSTM with the global contextual awareness of a Transformer. This significantly outperforms standalone LSTMs in capturing both short-term volatility and long-term momentum.

5. Branch D: The Relational Branch (Sector Correlation)

- **The Final Model:** SM2PNet / TRNet (Temporal Relational Network using GNNs).
- **How it works:** Stocks do not move in isolation. This branch constructs an adjacency matrix using Pearson's correlation coefficient to map the linkages between peer companies within the NIFTY 50.
- **Application:** When we are trading Bank Nifty, this model ingests the real-time movements of highly correlated peers (e.g., HDFC, ICICI, Kotak) to predict the movement of the broader index or a specific target firm.

6. Branch E: The Sentiment & Risk Branch (Imbalanced Data)

- **The Final Model:** Off-Policy PPO (Proximal Policy Optimization) + FinBERT.
- **How it works:** To analyse financial news or social media, we will use FinBERT, an LLM trained strictly on financial text.
- **Addressing Imbalanced Data:** Financial news suffers from "imbalanced classification"—it is mostly neutral/positive and rarely signals a crash.
- **Off-Policy PPO:** We will use this reinforcement learning algorithm to adjust the reward mechanism during training. It assigns a higher "reward" for correctly classifying the rare minority class (e.g., sudden negative news), preventing our model from lazily predicting that the market is always "neutral".
- **Combo Loss (CL):** We will train this branch using a Combo Loss function, which amplifies the impact of complex, rare samples. This vastly outperforms standard cross-entropy functions on skewed financial datasets.